Converting Second-Order Linear ODEs to Systems of First-Order Equations

A Comprehensive Lesson and Practice Set

The Core Concept

When dealing with differential equations, high-order derivatives can complicate both theoretical analysis and numerical computation. By introducing new variables, we can transform any single higher-order differential equation into a system of first-order equations—a powerful structural simplification.

We begin with the general form of a second-order linear differential equation:

$$a(t)x'' + b(t)x' + c(t)x = g(t) \tag{1}$$

Our goal is to eliminate the second derivative, x'', by defining a new set of state variables.

The Step-by-Step Conversion

Step 1: Define the state variables.
Let our first variable represent the original function, and our second variable represent its first derivative.

$$x_1(t) = x(t)$$
$$x_2(t) = x'(t)$$

Note that by the definition above, letting $v(t) = x'(t)$ is equivalent to establishing $x_2(t) = v(t)$. Therefore, taking the derivative of x_1 gives us our first equation of the system:

$$x_1' = x_2 \tag{2}$$

Step 2: Differentiate the second state variable.
Taking the derivative of x_2 yields the second derivative of our original function:

$$x_2' = x''$$

Step 3: Solve the original ODE for x''.
We return to our original equation and isolate the highest derivative, x'':

$$a(t)x'' = -b(t)x' - c(t)x + g(t)$$

$$x'' = -\frac{c(t)}{a(t)}x - \frac{b(t)}{a(t)}x' + \frac{g(t)}{a(t)}$$

Step 4: Substitute the state variables.
Replace x with x_1, x' with x_2, and x'' with x_2'. This yields our second equation:

$$x_2' = -\frac{c(t)}{a(t)}x_1 - \frac{b(t)}{a(t)}x_2 + \frac{g(t)}{a(t)} \tag{3}$$

Step 5: Write out the final system.
Combining equations (2) and (3) gives us the complete system of first-order linear equations:

System of First-Order Equations

$$
\begin{aligned}
x_1' &= x_2 \\
x_2' &= -\frac{c(t)}{a(t)}x_1 - \frac{b(t)}{a(t)}x_2 + \frac{g(t)}{a(t)}
\end{aligned}
$$

This system is entirely equivalent to the original second-order equation—it just distributes the complexity across multiple dimensions rather than a single high-order derivative.

Practice Problems

Convert the following second-order linear differential equations into a system of two first-order equations. Assume $x_1 = x$ and $x_2 = x'$.

1. $x'' + 5x' + 6x = 0$
2. $x'' - 3x' + 2x = \sin(t)$
3. $2x'' + 8x' - 4x = 10$
4. $t^2x'' + tx' + x = 0$
5. $x'' + \omega^2 x = 0 \quad$ (where ω is a constant)
6. $x'' - 2tx' + 4x = 0$
7. $5x'' - x = e^{2t}$
8. $x'' + 7x' = t^3$
9. $e^t x'' + x' - x = \cos(t)$
10. $(1 - t^2)x'' - 2tx' + 2x = 0$

Solutions

1. **Original:** $x'' + 5x' + 6x = 0$
   Isolate x'': $x'' = -6x - 5x'$
   System:

$$\begin{aligned} x_1' &= x_2 \\ x_2' &= -6x_1 - 5x_2 \end{aligned}$$

2. **Original:** $x'' - 3x' + 2x = \sin(t)$
   Isolate x'': $x'' = -2x + 3x' + \sin(t)$
   System:

$$\begin{aligned} x_1' &= x_2 \\ x_2' &= -2x_1 + 3x_2 + \sin(t) \end{aligned}$$

3. **Original:** $2x'' + 8x' - 4x = 10$
   Divide by 2: $x'' + 4x' - 2x = 5$
   Isolate x'': $x'' = 2x - 4x' + 5$
   System:

$$\begin{aligned} x_1' &= x_2 \\ x_2' &= 2x_1 - 4x_2 + 5 \end{aligned}$$

4. **Original:** $t^2x'' + tx' + x = 0$
   Isolate x'': $x'' = -\frac{1}{t^2}x - \frac{1}{t}x'$
   System:

$$x_1' = x_2$$
$$x_2' = -\frac{1}{t^2}x_1 - \frac{1}{t}x_2$$

5. **Original:** $x'' + \omega^2 x = 0$
   Isolate x'': $x'' = -\omega^2 x$
   System:

$$x_1' = x_2$$
$$x_2' = -\omega^2 x_1$$

6. **Original:** $x'' - 2tx' + 4x = 0$
   Isolate x'': $x'' = -4x + 2tx'$
   System:

$$x_1' = x_2$$
$$x_2' = -4x_1 + 2tx_2$$

7. **Original:** $5x'' - x = e^{2t}$
   Isolate x'': $x'' = \frac{1}{5}x + \frac{1}{5}e^{2t}$
   System:

$$x_1' = x_2$$
$$x_2' = \frac{1}{5}x_1 + \frac{1}{5}e^{2t}$$

8. **Original:** $x'' + 7x' = t^3$
   Isolate x'': $x'' = -7x' + t^3$
   System:

$$x_1' = x_2$$
$$x_2' = -7x_2 + t^3$$

9. **Original:** $e^t x'' + x' - x = \cos(t)$
   Isolate x'': $x'' = e^{-t}x - e^{-t}x' + e^{-t}\cos(t)$
   System:

$$x_1' = x_2$$
$$x_2' = e^{-t}x_1 - e^{-t}x_2 + e^{-t}\cos(t)$$

10. **Original:** $(1-t^2)x'' - 2tx' + 2x = 0$
    Isolate x'': $x'' = -\frac{2}{1-t^2}x + \frac{2t}{1-t^2}x'$
    System:

$$x_1' = x_2$$
$$x_2' = -\frac{2}{1-t^2}x_1 + \frac{2t}{1-t^2}x_2$$